Gaussian & Markov Processes

Lecture 8

Nov. 11, 13 and 18, 2025

Outline

Gaussian Random Process

- A Gaussian random process is a random process $X(t)$, $t \in \mathcal{T}$, such that
$$\begin{bmatrix} X(t_1) \\ X(t_2) \\ \vdots \\ X(t_n) \end{bmatrix}$$
is a Gaussian random vector for any $n \geq 1$ and all time instants $t_1, t_2, \cdots, t_n \in \mathcal{T}$.
- Since the joint pdf for a Gaussian random vector is specified by its mean and covariance matrix, a Gaussian random process is specified by its mean $\mu_X(t)$ and autocorrelation $R_X(t_1, t_2)$ functions.
- A WSS Gaussian random process is also SSS.

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  Suppose $X(t)$ is WSS (this requires $\mathcal{T} = (-\infty, \infty)$). Then $\mu_X(t) = m$ and $R_X(t_1, t_2) = R_X(|t_1 - t_2|)$. Now consider arbitrary $k \geq 1$, k time instants $t_1, t_2, \cdots, t_k$, and time shift τ. Since $X(t)$ is a Gaussian process, the vector

$$\boldsymbol{X} = [X(t_1)\ X(t_2)\ \cdots\ X(t_k)]^\top \sim \mathcal{N}(\boldsymbol{m}, \boldsymbol{\Sigma})$$

  where $\boldsymbol{m} = E\{\boldsymbol{X}\}$, $\boldsymbol{\Sigma} = \boldsymbol{R} - \boldsymbol{m}\boldsymbol{m}^\top$, $\boldsymbol{R} = E\{\boldsymbol{X}\boldsymbol{X}^\top\}$, with (i,j)th element $\boldsymbol{R}_{ij} = E\{X(t_i)X(t_j)\}$. Since $X(t)$ is WSS,

$$\boldsymbol{R}_{ij} = E\{X(t_i)X(t_j)\} = E\{X(t_i + \tau)X(t_j + \tau)\}$$

  for every τ and $\boldsymbol{m}$ is not a function of t. Hence, $\boldsymbol{X}' = [X(t_1 + \tau)\ X(t_2 + \tau)\ \cdots\ X(t_k + \tau)]^\top \sim \mathcal{N}(\boldsymbol{m}, \boldsymbol{\Sigma})$, i.e.,

$$X(t_1), X(t_2), \cdots, X(t_k) \text{ and } X(t_1 + \tau), X(t_2 + \tau), \cdots, X(t_k + \tau)$$

  have the same joint pdf. So Gaussian $X(t)$ is SSS if it is WSS.

- Any linear (affine) transformation of a Gaussian random process results in a Gaussian random process. This includes linear filtering:

$$Y(t) = \int_{-\infty}^{\infty} h(t-\tau)X(\tau)\,d\tau$$

or (time-varying)

$$Y(t) = \int_{-\infty}^{\infty} h(t, t-\tau)X(\tau)\,d\tau$$

- A proof is beyond the scope of this course. But this fact is important.
- Mean $\mu_X(t)$ and autocorrelation $R_X(t_1, t_2)$ functions of Gaussian $X(t)$ completely characterize the random process. If $Y(t)$ is the result of linear filtering of Gaussian $X(t)$, then we only need to know $\mu_Y(t)$ and autocorrelation $R_Y(t_1, t_2)$ which can be computed from $h(t)$ (or $h(t, t-\tau)$), $\mu_X(t)$ and $R_X(t_1, t_2)$, in general, and from $h(t)$ μ_X and $R_X(\tau)$ for WSS $X(t)$ (hence, WSS $Y(t)$).

Gauss-Markov Process

- Let Z_n, $n \geq 1$, be a WGN (White Gaussian Noise) process, i.e., an IID process with $Z_1 \sim \mathcal{N}(0, N)$. The Gauss-Markov process is a first-order autoregressive process defined by

$$\begin{aligned} X_1 &= Z_1 \\ X_n &= \alpha X_{n-1} + Z_n \quad n \geq 1\,, \end{aligned}$$

  where $|\alpha| < 1$.

- This process is a Gaussian random process since

$$\begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \\ \alpha & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \alpha^{n-2} & \alpha^{n-3} & \cdots & 1 & 0 \\ \alpha^{n-1} & \alpha^{n-2} & \cdots & \alpha & 1 \end{bmatrix} \begin{bmatrix} Z_1 \\ Z_2 \\ \vdots \\ Z_n \end{bmatrix}$$

  is a linear transformation of a Gaussian random vector and is therefore a Gaussian random vector.

- The Gauss-Markov process is Markov. It is not, however, an independent increment process.
  Recall that if the RVs $\{X_n : n \geq 1\}$ are continuous, then the process is Markov iff the pdfs

$$f_{X_{n+1}|\boldsymbol{X}^n}(x_{n+1}|x_n, \boldsymbol{x}^{n-1}) = f_{X_{n+1}|X_n}(x_{n+1}|x_n)$$

  for every n where $\boldsymbol{X}^n = \{X_i : 1 \leq i \leq n\}$. We have

$$\begin{aligned} f_{X_{n+1}|\boldsymbol{X}^n}(x_{n+1}|x_n, \boldsymbol{x}^{n-1}) &= f_{Z_n}(x_{n+1} - \alpha x_n) \\ &= f_{X_{n+1}|X_n}(x_{n+1}|x_n) \end{aligned}$$

- Mean function:

$$\begin{aligned} \mu_X(n) &= E\{X_n\} = E\{\alpha X_{n-1} + Z_n\} \\ &= \alpha E\{X_{n-1}\} + \underbrace{E\{Z_n\}}_{0} = \alpha E\{X_{n-1}\} \\ &= \cdots = \alpha^{n-1} E\{Z_1\} = 0 \end{aligned}$$

- covariance function: To find the autocorrelation function, for $n_2 > n_1$ we write

$$X_{n_2} = \alpha^{n_2-n_1} X_{n_1} + \sum_{i=1}^{n_2-n_1-1} \alpha^i Z_{n_2-i} \, .$$

Thus

$$R_X(n_1, n_2) = E\{X_{n_1} X_{n_2}\} = \alpha^{n_2-n_1} R_X(n_1, n_1) + 0 = \alpha^{n_2-n_1} E\{X_{n_1}^2\}$$

since, for $0 \leq i \leq n_2 - n_1 - 1$, X_{n_1} and Z_{n_2-i} are independent, zero mean.

Next, to find $E\{X_{n_1}^2\}$, consider

$$\begin{aligned} E\{X_1^2\} &= E\{Z_1^2\} = N \\ E\{X_{n_1}^2\} &= E\{(\alpha X_{n_1-1} + Z_{n_1})^2\} = \alpha^2 E\{X_{n_1-1}^2\} + E\{Z_{n_1}^2\} + 0 \\ &= \alpha^2 E\{X_{n_1-1}^2\} + N \end{aligned}$$

Using the above recursion, we have

$$E\{X_{n_1}^2\} = \frac{1-\alpha^{2n_1}}{1-\alpha^2} N$$

- Let $y_n = \beta y_{n-1} + N$ with $y_1 = N$. Then we have

$$
\begin{aligned}
y_2 &= \beta y_1 + N = \beta N + N \\
y_3 &= \beta y_2 + N = \beta(\beta y_1 + N) + N = \beta^2 N + \beta N + N \\
\vdots &= \vdots \\
y_n &= N \sum_{i=0}^{n-1} \beta^i = N\Big[\sum_{i=0}^{\infty} \beta^i - \sum_{i=n}^{\infty} \beta^i\Big] \\
&= N\Big[\frac{1}{1-\beta} - \beta^n \sum_{\ell=0}^{\infty} \beta^\ell\Big] = N \frac{1-\beta^n}{1-\beta}
\end{aligned}
$$

Apply the above with $\beta = \alpha^2$ and $n = n_1$ to obtain the result

$$E\{X_{n_1}^2\} = \frac{1-\alpha^{2n_1}}{1-\alpha^2} N$$

- Finally the autocorrelation function is

$$R_X(n_1, n_2) = \alpha^{|n_2 - n_1|} \, \frac{1-\alpha^{2\min(n_2, n_1)}}{1-\alpha^2} \, N$$

Wiener Process

- The Wiener process $W(t)$, $t \geq 0$, is an independent increment process such that
  - $W(0) = 0$
  - $W(t_2) - W(t_1) \sim \mathcal{N}(0, \sigma^2(t_2 - t_1))$ for all $t_2 > t_1 \geq 0$
  - The sample functions $W(t, \omega)$ are continuous in t.

  If $\sigma^2 = 1$, it is called standard Wiener process.
- Consider a random walk with step size s (before we had $s = 1$), a new step taken every T sec:

$$X_n = s \sum_{i=1}^{n} Z_i \to \underbrace{X_T(nT)}_{x_T(t)} = s \sum_{i=1}^{\infty} Z_i u(t - iT)$$

$$E\{x_T(nT)\} = 0\,, \quad E\{x_T^2(nT)\} = s^2 n = s^2 \frac{t}{T}\,.$$

Let $T \to 0$, $s \to 0$ and $n \to \infty$ such that $t = nT$ =constant and $s^2 \frac{t}{T}$ is finite:

$$X_T(nT) \to X(t) = W(t) \,:\, E\{x_T^2(nT)\} = \sigma^2 t$$

- For a Wiener process $W(t)$,

$$E\{W(t)\} = 0 \text{ and } E\{W^2(t)\} = \sigma^2 t$$

- The first-order pdf of $W(t_2) - W(t_1)$, $t_2 > t_1 \geq 0$, is

$$f_{W(t_2)-W(t_1)}(w) = \frac{1}{\sqrt{2\pi\sigma^2(t_2 - t_1)}} e^{-\frac{w^2}{2\sigma^2(t_2-t_1)}}$$

- Autocorrelation Function:

$$\begin{aligned}
R_W(t_1, t_2) &= E\{W(t_1)W(t_2)\} \\
&\overset{\text{if } t_2 \geq t_1}{=} E\{W(t_1)(W(t_2) - W(t_1) + W(t_1))\} \\
&= E\{W(t_1)\}E\{(W(t_2) - W(t_1))\} + E\{W^2(t_1)\} \\
&= 0 + \sigma^2 t_1 = \sigma^2 t_1
\end{aligned}$$

Considering both cases, $t_2 \geq t_1$ and $t_2 < t_1$, we have

$$R_W(t_1, t_2) = \sigma^2 \min(t_1, t_2)$$

- The second-order pdf of Wiener process is

$$
\begin{aligned}
f_{W(t_2),W(t_1)}(w_1, w_2) &= f_{W(t_2)-W(t_1),W(t_1)}(w_2 - w_1, w_1) \qquad (1)\\
&= \frac{1}{\sqrt{2\pi\sigma^2(t_2 - t_1)}} e^{-\frac{(w_2-w_1)^2}{2\sigma^2(t_2-t_1)}} \frac{1}{\sqrt{2\pi\sigma^2 t_1}} e^{-\frac{w_1^2}{2\sigma^2 t_1}}
\end{aligned}
$$

The second equality above follows from the independent increment property of $W(t)$. To see the first equality, consider the transformation

$$
\begin{bmatrix} Y \\ Z \end{bmatrix} = \begin{bmatrix} W_2 \\ W_1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} W_2 - W_1 \\ W_1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} U \\ V \end{bmatrix}
$$

Then there is a single solution for $u = y - z$ and $v = z$ given (y, z), and the determinant of the Jacobian is one. So we have

$$
f_{YZ}(y, z) = f_{UV}(y - z, z) \Rightarrow \text{Equation (1)}
$$

- The independent increment property of $W(t)$ allows one to calculate the joint pdf of any order.

Markov Chains

- A discrete-time Markov process $X_0, X_1, X_2, \cdots$ is called a Markov Chain if
  - For all $n \geq 0$, $X_n \in \mathcal{S}$, where $\mathcal{S}$ is a finite set called the state space. We often assume that $\mathcal{S} = \{1, 2, \cdots, m\}$, a finite set indexed by positive integers. But $\mathcal{S}$ can be countably infinite.
  - For $n \geq 0$ and $i, j \in \mathcal{S}$,

    $$P(X_{n+1} = j | X_n = i) = p_{ij}\ , \text{ independent of } n$$

    This class is called homogeneous Markov chain. If $P(X_{n+1} = j | X_n = i)$ depends on n, then it is called a non-homogeneous Markov chain.
- So, a homogeneous Markov chain is specified by a transition probability matrix (for $\mathcal{S} = \{1, 2, \cdots, m\}$)

$$\mathcal{P} = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1m} \\ p_{21} & p_{22} & \cdots & p_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ p_{m1} & p_{m2} & \cdots & p_{mm} \end{bmatrix}$$

Clearly $\sum_{j=1}^{m} p_{ij} = 1$ for all i, i.e., the sum of any row is 1

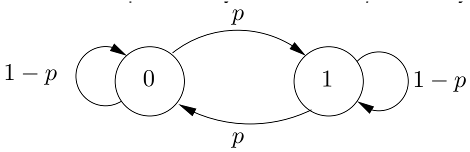
- By the Markov property, for all $n \geq 0$ and all states

$$P(X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \cdots, X_0 = i_0) = P(X_{n+1} = j | X_n = i) = p_{ij}$$

- Example: Binary Symmetric Markov Chain: Consider a sequence of coin flips, where each flip has probability of $1 - p$ of having the same outcome as the previous coin flip, regardless of all previous flips. The probability transition matrix (head is 1 and tail is 0) is

$$\mathcal{P} = \begin{bmatrix} 1-p & p \\ p & 1-p \end{bmatrix}$$

A Markov chain can are be specified by a transition probability graph (also state transition diagram)



Nodes are states (i.e., values), arcs are state transitions; (i, j) from state i to state j (only draw transitions with $p_{ij} > 0$)

- Can construct this process from an IID process: Let $Z_1, Z_2, \cdots$ be a Bernoulli process with parameter p. The binary symmetric Markov chain X_n can be defined as

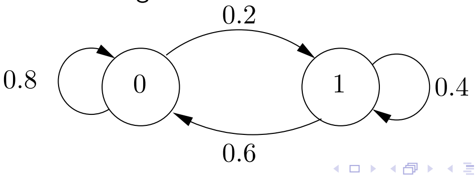
$$X_{n+1} = (X_n + Z_n) \bmod 2, \text{ for } n = 1, 2, \cdots$$

  So, each transition corresponds to passing the RV X_n through a binary symmetric channel with additive noise $Z_n \sim \text{Bernoulli}(p)$

- Example: Asymmetric Binary Markov Chain: State 0: Machine is working, State 1: Machine is broken down.
  The probability transition matrix is

$$\mathcal{P} = \begin{bmatrix} 0.8 & 0.2 \\ 0.6 & 0.4 \end{bmatrix}$$

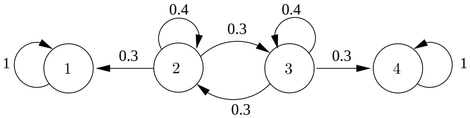
  and the state transition diagram is



- Example: Two spiders and a fly: A fly's possible positions are represented by four states:
  States 2,3 : safely flying in the left or right half of a room
  State 1 : A spider's web on the left wall
  State 4 : A spider's web on the right wall
  The probability transition matrix is

$$\mathcal{P} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.3 & 0.4 & 0.3 & 0 \\ 0 & 0.3 & 0.4 & 0.3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

and the state transition diagram is



- Given a Markov chain model, we can compute the probability of a sequence of states given an initial state $X_0 = i_0$ using the chain rule:

$$
\begin{aligned}
&P(X_1 = i_1, X_2 = i_2, \cdots, X_n = i_n | X_0 = i_0) \\
&\quad = P(X_1 = i_1 | X_0 = i_0) P(X_2 = i_2 | X_0 = i_0, X_1 = i_1) \cdots \\
&\qquad\qquad P(X_n = i_n | X_0 = i_0, X_1 = i_1, \cdots, X_{n-1} = i_{n-1}) \\
&\quad = P(X_1 = i_1 | X_0 = i_0) P(X_2 = i_2 | X_1 = i_1) \cdots P(X_n = i_n | X_{n-1} = i_{n-1}) \\
&\quad = p_{i_0 i_1} p_{i_1 i_2} \cdots p_{i_{n-1} i_n}
\end{aligned}
$$

- Initial state probability is needed for unconditional probability:

$$
\begin{aligned}
&P(X_0 = i_0, X_1 = i_1, X_2 = i_2, \cdots, X_n = i_n) \\
&\quad = P(X_0 = i_0) P(X_1 = i_1, X_2 = i_2, \cdots, X_n = i_n | X_0 = i_0) \\
&\quad = P(X_0 = i_0) p_{i_0 i_1} p_{i_1 i_2} \cdots p_{i_{n-1} i_n}
\end{aligned}
$$

- Example: For the spider and fly example,

$$
P(X_1 = 2, X_2 = 2, X_3 = 3 | X_0 = 2) = p_{22} p_{22} p_{23} = 0.4 \times 0.4 \times 0.3 = 0.048
$$

- There are many other questions of interest, including:
  - n-state transition probabilities: Beginning from some state i what is the probability that in n steps we end up in state j?
  - Steady state probabilities: What is the expected fraction of time spent in state i as $n \to \infty$?
- Define the n-step transition probabilities as

$$p_{ij}(n) = P(X_n = j \,|\, X_0 = i)\,, \text{ for } 1 \le i, j \le m \text{ (= no. of states)}$$

- The n-step transition probabilities can be computed using the Chapman-Kolmogorov recursive equation:

$$p_{ij}(n) = \sum_{k=1}^{m} p_{ik}(n-1) p_{kj} \text{ for } n > 1, \text{ and all } 1 \le i, j \le m\,,$$

starting with $p_{ij}(1) = p_{ij}$. By the law of total probability

$$P(X_n = j \,|\, X_0 = i) = \sum_k P(X_n = j, X_{n-1} = k \,|\, X_0 = i)$$

$$= \sum_k P(X_n = j \,|\, X_{n-1} = k, X_0 = i) P(X_{n-1} = k \,|\, X_0 = i)$$

$$= \sum_k P(X_{n-1} = k \,|\, X_0 = i) P(X_n = j \,|\, X_{n-1} = k) = \sum_k p_{ik}(n-1) p_{kj}$$

- We can view the $p_{ij}(n)$, $1 \leq i, j \leq m$, as the elements of a matrix $\mathcal{P}(n)$, called the n-step transition probability matrix. Then we can view the Kolmogorov-Chapman equations as a sequence of matrix multiplications:

$$\begin{aligned} \mathcal{P}(1) &= \mathcal{P} \\ \mathcal{P}(2) &= \mathcal{P}(1)\mathcal{P} = \mathcal{P}^2 \\ \vdots &= \vdots \\ \mathcal{P}(n) &= \mathcal{P}(n-1)\mathcal{P} = \mathcal{P}^n \end{aligned}$$

- Example: For the binary asymmetric Markov chain

$$\mathcal{P}(1) = \begin{bmatrix} 0.8 & 0.2 \\ 0.6 & 0.4 \end{bmatrix}, \quad \mathcal{P}(2) = \mathcal{P}^2 = \begin{bmatrix} 0.76 & 0.24 \\ 0.72 & 0.28 \end{bmatrix}$$

$$\mathcal{P}(3) = \mathcal{P}^3 = \begin{bmatrix} 0.752 & 0.248 \\ 0.744 & 0.256 \end{bmatrix}, \quad \lim_{n\to\infty} \mathcal{P}(n) = \lim_{n\to\infty} \mathcal{P}^n = \begin{bmatrix} 0.75 & 0.25 \\ 0.75 & 0.25 \end{bmatrix}$$

- In this example (binary asymmetric Markov chain), each $p_{ij}(n)$ converges to a non-zero limit independent of the initial state (i), i.e., each state (j) has a steady state probability of being occupied as $n \to \infty$.
- Example: Consider the spiders-and-fly example

$$\mathcal{P}(1) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.3 & 0.4 & 0.3 & 0 \\ 0 & 0.3 & 0.4 & 0.3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathcal{P}(2) = \mathcal{P}^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.42 & 0.25 & 0.24 & 0.09 \\ 0.09 & 0.24 & 0.25 & 0.42 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Continuing with the spiders-and-fly example

$$\mathcal{P}(3) = \mathcal{P}^3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.50 & 0.17 & 0.17 & 0.16 \\ 0.16 & 0.17 & 0.17 & 0.50 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\lim_{n\to\infty} \mathcal{P}(n) = \lim_{n\to\infty} \mathcal{P}^n = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2/3 & 0 & 0 & 1/3 \\ 1/3 & 0 & 0 & 2/3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Here $p_{ij}(n)$ converges, but the limit depends on the initial state and can be 0 for some states. Note that states 1 and 4 corresponding to capturing the fly by one of the spiders are absorbing states, i.e., they are infinitely repeated once visited. The probability of being in non-absorbing states 2 and 3 diminishes as time increases.

- Define $p_i(n) = P(X_n = i)$= probability that the system is in state i at time n. Let

$$\boldsymbol{p}(n) = \begin{bmatrix} p_1(n) & p_2(n) & \cdots & p_m(n) \end{bmatrix} \in \mathbb{R}^{1\times m}.$$

- For any $k < n$ we have (law of total probability)

$$\begin{aligned} p_j(n) &= \sum_{i=1}^{m} P(X_n = j, X_k = i) \\ &= \sum_{i=1}^{m} P(X_n = j | X_k = i) P(X_k = i) = \sum_{i=1}^{m} p_{ij}(n-k) p_i(k) \end{aligned}$$

$$\boldsymbol{p}(n) = \boldsymbol{p}(k)\mathcal{P}^{n-k} = \boldsymbol{p}(0)\mathcal{P}^{n}$$

- Steady-State Probability Distribution: If it exists (i.e., $\lim_{n\to\infty} p_{ij}(n) = \lim_{n\to\infty} p_j(n) = \pi_j$ for all i, and the limits are unique),

$$\lim_{n\to\infty} \boldsymbol{p}(n) = \boldsymbol{\Pi} = \begin{bmatrix} \pi_1 & \pi_2 & \cdots & \pi_m \end{bmatrix}$$

satisfying the balance equations

$$\boldsymbol{\Pi} = \boldsymbol{\Pi}\mathcal{P} \text{ subject to normalization equation } \sum_{i=1}^{m} \pi_i = 1.$$

- Steady-State Probability Distribution:

$$\boldsymbol{\Pi} = \boldsymbol{\Pi}\mathcal{P}$$

follows from the Chapman-Kolmogorov equations by letting $n \to \infty$

$$p_{ij}(n+1) = \sum_{k=1}^{m} p_{ik}(n)p_{kj} \text{ for } 1 \leq i,j \leq m\,.$$

- Example: Binary Symmetric Markov Chain

$$\mathcal{P} = \begin{bmatrix} 1-p & p \\ p & 1-p \end{bmatrix}$$

Find the steady-state probabilities.
Solution: We need to solve the balance and normalization equations.

- Balance and normalization equations:

$$\pi_1 = \pi_1 p_{11} + \pi_2 p_{21} = \pi_1(1-p) + \pi_2 p$$
$$\pi_2 = \pi_1 p_{12} + \pi_2 p_{22} = \pi_1 p + \pi_2(1-p)$$
$$1 = \pi_1 + \pi_2$$

  Note that the first two equations are linearly dependent. Both yield $\pi_1 = \pi_2$. Substituting in the last equation, we obtain $\pi_1 = \pi_2 = 0.5$

- Example: Asymmetric binary Markov chain

$$\mathcal{P} = \begin{bmatrix} 0.8 & 0.2 \\ 0.6 & 0.4 \end{bmatrix}$$

  The steady-state equations are

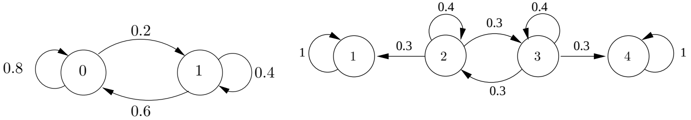
$$\pi_1 = \pi_1 p_{11} + \pi_2 p_{21} = 0.8\pi_1 + 0.2\pi_2 \text{ and } 1 = \pi_1 + \pi_2$$

  Solving the two equations yields $\pi_1 = 3/4$ and $\pi_2 = 1/4$.

Classification of States

- As we have seen, various states of a Markov chain can have different characteristics.
- We wish to classify the states by the long-term frequency with which they are visited.
- Let $A(i)$ be the set of states that are accessible from state i (may include i itself), i.e., can be reached from i in n steps, for some n.
- State i is said to be recurrent if starting from i, any accessible state j must be such that i is accessible from j, i.e., $j \in A(i)$ iff $i \in A(j)$. Clearly, this implies that if i is recurrent then it must be in $A(i)$.
- A state is said to be transient if it is not recurrent.
- Note that recurrence/transcience is determined by the arcs (transitions with nonzero probability), not by actual values of probabilities.

- Example: Classify the states of the following Markov chains:



  - Left Fig: States 0 and 1 are recurrent
  - Right Fig: States 1 and 4 are recurrent, and States 2 and 3 are transient
- The set of accessible states $A(i)$ from some recurrent state i is called a recurrent class.
- Every state k in a recurrent class $A(i)$ is recurrent and $A(k) = A(i)$.
- Two recurrent classes are either identical or disjoint.
- Steady-state convergence theorem: If a Markov chain has only one recurrent class and it is not periodic, then $p_{ij}(n)$ tends to a steady-state π_j independent of i, i.e., $\lim_{n\to\infty} p_{ij}(n) = \pi_j$ for all i